

Shambhala-2: A Protocol for Uniformly Shaped Harmonization of Gene Expression Profiles of Various Formats

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Uniformly shaped harmonization of gene expression profiles is central for the simultaneous comparison of multiple gene expression datasets. It is expected to operate with the gene expression data obtained using various experimental methods and equipment, and to return harmonized profiles in a uniform shape. Such uniformly shaped expression profiles from different initial datasets can be further compared directly. However, current harmonization techniques have strong limitations that prevent their broad use for bioinformatic applications. They can either operate with only up to two datasets/platforms or return data in a dynamic format that will be different for every comparison under analysis. This also does not allow for adding new data to the previously harmonized dataset(s), which complicates the analysis and increases calculation costs. We propose here a new method termed Shambhala-2 that can transform multi-platform expression data into a universal format that is identical for all harmonizations made using this technique. Shambhala-2 is based on sample-by-sample cubic conversion of the initial expression dataset into a pre-selected shape of the reference definitive dataset. Using 8390 samples of 12 healthy human tissue types and 4086 samples of colorectal, kidney, and lung cancer tissues, we verified Shambhala-2's capacity in restoring tissue-specific expression patterns for seven microarray and three RNA sequencing platforms. Shambhala-2 performed well for all tested combinations of RNAseq and microarray profiles, and retained gene-expression ranks, as evidenced by high correlations between different single- or aggregated gene expression metrics in pre- and post-Shambhalized samples, including preserving cancer-specific gene expression and pathway activation features. © 2022 Wiley Periodicals LLC.

Basic Protocol: Shambhala-2 harmonizer

Alternate Protocol 1: Linear Shambhala/Shambhala-1

Alternate Protocol 2: Alternative (flexible-format and uniformly shaped) normalization methods

Support Protocol 1: Watermelon multisection (WM)

Support Protocol 2: Calculation of cancer-to-normal log-fold-change (LFC) and pathway activation level (PAL)

Keywords: data normalization and harmonization • gene expression • microarray hybridization • RNA sequencing • transcriptional profiles

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INTRODUCTION

List of Acronyms

AMCF	activation-to-mitosis conversion factor
ARR	activator/repressor role
DESeq2	differential gene expression analysis 2
GTE _x	Genotype Tissue Expression
LFC	log-fold-change
MH	microarray hybridization
NGS	next-generation sequencing
NII	node involvement index
PAL	pathway activation level
PC	principal component
PCA	principal component analysis
QN	quantile normalization
QNR	quantile normalization (robust)
RNAseq	RNA sequencing
Sh2	Shambhala-2
TCGA	The Cancer Genome Archive
TDM	Training Distribution Machine
UPC	Universal exPression Code
WM	watermelon multisection
XPN	cross-platform normalization

Shambhala-2

The number of publicly available gene expression profiles, and consequently the amount of transcriptomic information, grows exponentially. For example, in early 2019 there were about 70,000 datasets with roughly 2 million distinct gene expression profiles (samples) in the Gene Expression Omnibus (GEO) repository (Borisov et al., 2019), and in late 2021, these numbers increased to ~165,000 and ~4.8 million, respectively. The overwhelming majority of the samples were obtained using only two alternative and technologically different approaches: expression microarray hybridization (MH; Adomas et al., 2008; Lashkari et al., 1997) and RNA sequencing (RNAseq; Chu & Corey, 2012; Ingolia, Brar, Rouskin, McGeachy, & Weissman, 2012; Maher et al., 2009; Nagalakshmi et al., 2008; Wang, Gerstein, & Snyder, 2009). However, for each of these two approaches, different equipment platforms and reagent settings could be in use, which strongly affects final gene expression profiles.

The apparent technical heterogeneity of gene expression data essentially reduces our ability to compare multiple datasets at once, both at the cross-platform and at the intra-platform levels (Bartlett, Dhruva, Shah, Ryan, & Ross, 2019; Berger et al., 2017; Borisov et al., 2019). The latter is mostly due to the so-called *batch effect*, which frequently arises even within the same platform and reagent settings; however, similar effects can be seen with the comparison of different sets of experimental data (Aliper et al., 2017; Borisov et al., 2017, 2021; Demetrashvili, Kron, Pethe, Bapat, & Briollais, 2010).

To solve these problems, several analytical approaches were proposed at both the intra- and cross-platform levels. For intra-platform normalization methods, quantile

normalization (QN; Bolstad, Irizarry, Astrand, & Speed, 2003), frozen robust multi-array analysis for microarray hybridization data (fRMA; McCall, Bolstad, & Irizarry, 2010), the Empirical Bayes (EB) method, also known as ComBat (Walker et al., 2008), and Differential Expression analysis for Sequence count data (DESeq/DESeq2; Anders & Huber, 2010; Love, Huber, & Anders, 2014) were developed. In turn, the methods for cross-platform normalization include distance-weighted discrimination (DWD; Benito et al., 2004; Huang, Lu, Liu, Haaland, & Marron, 2012), cross-platform normalization (XPN; Rudy & Valafar, 2011; Shabalin, Tjelmeland, Fan, Perou, & Nobel, 2008), Quantile Discretization (Warnat, Eils, & Brors, 2005), Normalized Discretization (Warnat et al., 2005), Distribution Transformation (Jiang et al., 2004), Gene Quantiles (Xia et al., 2009), Platform-Independent Latent Dirichlet Allocation (PLIDA; Deshwar & Morris, 2014), Training Distribution Machine (TDM; Thompson, Tan, & Greene, 2016), Universal exPReSSion Code (UPC; Piccolo, Withers, Francis, Bild, & Johnson, 2013), and Rank-In (Tang et al., 2021) techniques. However, most of these methods simultaneously normalize expression data for all profiles under analysis, and the output data format strongly depends on the initial sample, which presents issues when initial samples have used different methods and new data is added to an existing dataset. Thus, when new profile(s) become available, the procedure must be repeated with a complete new set of expression data, which complicates the use of such normalization for large and growing datasets due to high computation time/costs. Furthermore, many such harmonization/normalization methods cannot sufficiently suppress batch effect and platform-specific bias, which remain serious obstacles for the comparison of different expression datasets (Borisov et al., 2017; Demetrashvili et al., 2010).

In 2019, we introduced an alternative approach termed Shambhala or Shambhala-1 (Borisov et al., 2019), which utilized a conceptually new rationale for transforming all the samples into a uniform/standard gene expression data format. This is an agnostic method for gene expression data harmonization, which theoretically can process an unlimited number of samples from an unlimited number of experimental platforms one-by-one. Shambhala-1 allows one-by-one adding of new harmonized expression samples to a common pre-calculated pool, and no recalculation is needed for the whole set of samples. This feature can dramatically reduce calculation time and costs, which is especially sensitive considering next-level gene expression metrics such as molecular pathway activation levels (Aliper et al., 2017; Borisov et al., 2017; Borisov, Sorokin, Garazha, & Buzdin, 2020; Buzdin, Prassolov, Zhavoronkov, & Borisov, 2017; Buzdin et al., 2014, 2018), individual drug sensitivity estimates (Poddubskaya et al., 2019; Tkachev, Sorokin, Garazha et al., 2020; Zolotovskaia et al., 2019), machine learning models (Borisov & Buzdin, 2019; Borisov et al., 2018, 2021; Tkachev et al., 2019; Tkachev, Sorokin, Borisov, et al., 2020), and more. Technically, the Shambhala-1 method utilizes a piecewise linear normalization method (XPN) for building a universal gene expression matrix (Shabalin et al., 2008). Shambhala-1's performance was tested by sample-type clustering on the model groups of samples experimentally profiled using different microarray and RNAseq platforms.

More recently, a new approach termed CuBlock (Junet, Farrés, Mas, & Daura, 2021) was published for cross-platform harmonization of microarray expression data. CuBlock performs harmonization using piecewise cubic transformation, and may essentially outperform Shambhala-1 in local comparisons (Junet et al., 2021). However, CuBlock transforms samples into a flexible and non-uniform shape/format, which differs for the different harmonization procedures.

In this protocol, we attempted to combine benefits of Shambhala-1's rationale with CuBlock's mathematical apparatus. We generated a new hybrid protocol, termed Shambhala-2, which utilizes independent, universally shaped (like in Shambhala-1), cubic cross-platform (like in CuBlock) harmonization for every expression profile. Using a

total of 8390 samples for 12 healthy human tissue types, we checked Shambhala-2's performance in restoring tissue-specific expression patterns for nine microarray and RNAseq platforms. In a series of functional tests, Shambhala-2 was found to be significantly more effective than Shambhala-1 for the comparison of both RNAseq and microarray profiles. Shambhala-2's capacity for removing platform bias was compared with that of the three previous methods with flexible output formats: QN, DESeq2, and CuBlock, as well as with the three normalization methods with a uniformly formatted output: the so-called robust QN (QNR; Bolstad, 2008), TDM (Thompson et al., 2016), and UPC (Piccolo et al., 2013). Assessment of clustering quality and tissue-specific data retention metrics reveal that Shambhala-2 is currently unparalleled for the expression data harmonization in a uniform format. Additionally, we checked the ability of Shambhala-2 to retain cancer-to-normal gene expression ratios in the RNAseq gene expression profiles from colorectal, kidney, and lung cancers. We also explored Shambhala-2's capacity to retain the respective pathway activation levels (Borisov et al., 2020) calculated for the same sets of cancer and normal tissue RNAseq profiles. In most comparisons, Shambhala-2 demonstrated reasonably high correlation levels between such expression metrics for the profiles before and after harmonization, and outperformed other available approaches such as DESeq2.

BASIC PROTOCOL

CUBIC SHAMBHALA/SHAMBHALA-2

We aimed to improve the previous method, Shambhala-1 (Borisov et al., 2019), preserving its ability to return the harmonized expression data in a universal intercomparable format, but also enhancing the method's performance. The core concept of Shambhala methods (Fig. 1) is the conversion of the gene expression profiles into the format of a selected, definitive reference dataset (Q) obtained using the selected and pre-adjusted reference platform. In its present form, the tool was tailored for the comparison of human gene expression data, and its application for the expression profiles of other organisms requires further adjustments.

Let us consider an arbitrary set of experimental gene expression platforms that produced a set of gene expression profiles. The goal of the Shambhala method is, simultaneously, to make such profiles technically comparable (e.g., like in flexible-output normalization instruments), but universally shaped so that they would all be compatible with a pre-defined reference gene expression sample (this procedure may be called uniformly shaped harmonization). This reference dataset may be built from a set of expression profiles obtained employing a trusted and widely used experimental platform; this set is called here the reference definitive dataset (Q). In the Shambhala-2 method, the profiles are transformed one by one. The process of each profile transformation includes several iteration steps, where the dataset containing expression profiles under harmonization (called here dataset P) is reshaped while dataset Q remains unchanged. Consequently, the final output of such transformation has gene expression profiles reshaped into the format of dataset Q .

In many previous approaches, the size of datasets under harmonization can strongly bias the output results. To ensure comparable harmonization results for the datasets of different/unbalanced size, a specific step of expression data equalization is introduced for datasets from different platforms or that are structurally different. This equalization step uses another pre-defined dataset, the auxiliary calibration dataset (P), to make the gene expression distribution comparable for each individual profile before transformation into the Q -shape. In the other words, this means that each individual gene expression profile under harmonization, say i , is transformed into the Q -shape not within the initial dataset of unharmonized profiles, but only when it is taken alone, and afterwards in a transiently transformed shape it is merged with the dataset P .

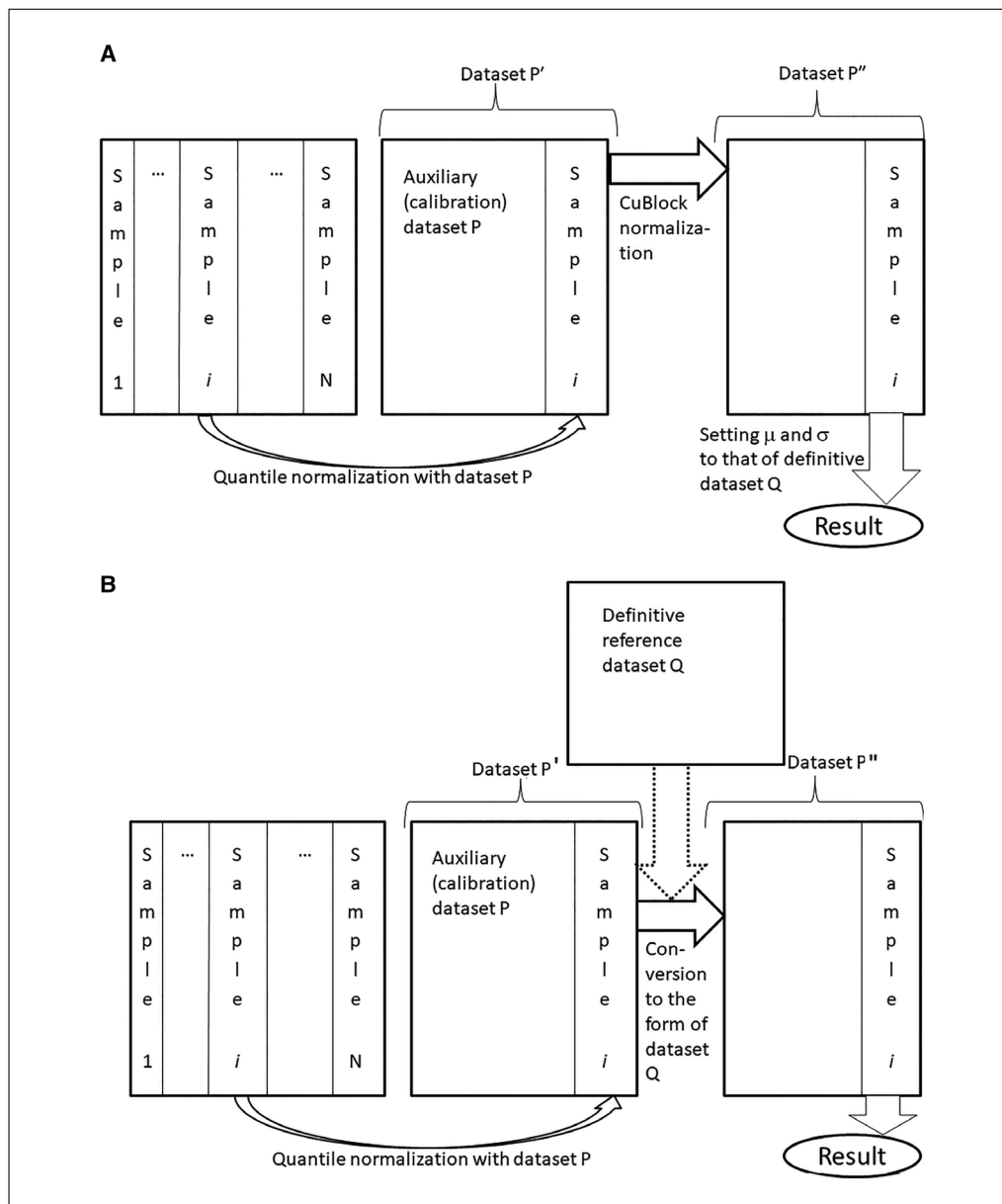


Figure 1 Outline of Shambhala approach for uniformly shaped harmonization of gene expression data. Gene expression profiles for samples (1,..., N), for example, obtained using different experimental platforms, are taken one-by-one, separately merged, and quantile-normalized with an *auxiliary calibration dataset P*. The resulting dataset P' is then transformed into the format of the *reference definitive dataset Q*, which results in *dataset P''*. The latter contains the finally transformed profile of initial sample i , which is considered harmonized. The profiles of all other samples are harmonized one-by-one using the same algorithm. **(A)** Shambhala-2 cubic harmonization protocol based on CuBlock method (Junet et al., 2021); **(B)** Shambhala-1 linear harmonization protocol (Borisov et al., 2019) based on XPN method (Shabalin et al., 2008).

Specifically, we quantile-normalize (Bolstad et al., 2003) profile i with the dataset P , which produces the dataset P' for further transformation. This dataset P' is then transformed into the shape of the dataset Q , thus producing the dataset P'' . From this dataset P'' , only one transformed profile i'' is taken for further analysis. This procedure is then applied to all other gene expression profiles under analysis from any number of starting datasets (Fig. 1A and B).

In Shambhala-2 harmonization, cubic normalization is used based on the CuBlock method (Junet et al., 2021). The dataset P' obtained after quantile normalization of the

Table 1 Expression Data Statistics for Harmonization Tests

Expression data platform	Sample type/number of samples				
	Test 1 (RNAseq-only)				
	Adrenal gland	Brain	Kidney	Liver	Thyroid
RNAseq; GTEx, Illumina HiSeq 2000, GPL11154	190	1671	45	175	446
RNAseq; TCGA, Illumina HiSeq 2000, GPL11154	3	5	129	50	59
RNAseq; Oncobox ANTE, Illumina HiSeq 3000, GPL21493	5	5	6	6	6
	Test 2 (RNAseq versus MH)				
	Brain	Fat	Muscle	Skin	Thyroid
RNAseq; GTEx, Illumina HiSeq 2000, GPL11154	1671	797	564	1203	446
MH; GTEx, Affymetrix HUG1, GPL16977	206	33	86	65	69
	Test 3 (MH only)				
	A (Stratagene RNA mix without brain)	B (Ambion brain RNA)	C (A:B mix 3:1)	D (A:B mix 1:3)	Brain
MH; MAQC, Agilent-012391 Whole Human Genome Oligo Microarray G4112A, GPL1708	15	15	14	15	-
MH; MAQC, Affymetrix Human Genome U133 Plus 2.0 Array, GPL570	15	15	14	15	-
MH; Illumina Sentrix Human-6 Expression Beadchip, GPL2507	15	15	14	15	-
MH; SEQC, Illumina HT-12, GPL10558	6	6	6	6	-
MH; SEQC, Affymetrix HUG2, GPL17930	4	4	4	4	-
MH; SEQC, Affymetrix PRV, GPL16043	4	4	4	4	-
MH; TCGA, Affymetrix U133, GPL570	-	-	-	-	10
MH; GTEx, Affymetrix HUG1, GPL16977	-	-	-	-	206

dataset P with added individual gene expression profile i under harmonization is then normalized using CuBlock method. From the CuBlock output dataset (P'), only one profile I' is taken for further operations. This profile is then rescaled to adjust gene expression values of each gene g in accordance with the mean expression and the standard expression deviation values of the definitive dataset Q equal to μ_{gQ} and σ_{gQ} , respectively (Fig. 1A).

Shambhala-2 was performed in R using MATLAB code and is publicly available at <https://github.com/BorisovNM/Shambhala2>.

Materials

Selection of expression datasets for harmonization tests:

For each type of harmonization test, i.e., between RNAseq data, RNAseq versus microarray hybridization (MH), and between MH data, we specifically selected five types of different human tissue samples, which showed clear-cut clustering pattern on dendrograms for each source (non-harmonized) dataset (Table 1).

For the RNAseq-only test we used human expression profiles from the Genotype Tissue Expression (GTEx; GTEx Consortium, 2013), The Cancer Genome Atlas (TCGA; Tomczak, Czerwińska, & Wiznerowicz, 2015), and Oncobox Atlas of

Table 2 Overview of Shambhala *Q*- and *P*-Datasets

Dataset/ Version	Platform	GPL code	Number of samples
Q_0	GTEx Illumina HiSeq 2000	GPL11154	100
Q_1	GTEx Affymetrix Human Gene 1.1 ST Array	GPL16977	100
P_0	Affymetrix Human Genome U133A 2.0 Array	GPL570	39
P_1	Agilent microarray	GPL4133 + GPL1708	43
P_2	CustomArray electrochemical microchip (CustomArray Inc., USA)	No special code	39
P_3	Illumina HumanHT-12 V4.0 expression beadchip	GPL10558	38
P_4	Quantile-normalized weighted mix of 50% of dataset P_3 + 50% of dataset P_1	GPL10558 + GPL4133 + GPL1708	81
P_5	Quantile-normalized weighted mix of 75% of dataset P_3 + 25% of dataset P_1	GPL10558 + GPL4133 + GPL1708	60
P_6	Illumina HiSeq 3000	GPL21493	36
P_7	Illumina HiSeq 2000	GPL11154	41

Normal Tissue Expression (ANTE; Suntsova et al., 2019) datasets. For each dataset, we took normal (non-cancer) profiles for adrenal gland, brain, kidney, liver, and thyroid tissues. All these profiles were obtained with Illumina engines HiSeq2000 and 3000, but using different protocols and reagents for each dataset.

In the RNAseq versus MH test, one microarray platform (Affymetrix HUG1) and one RNAseq platform (Illumina HiSeq2000) were compared, for which we extracted expression data for healthy brain, fat, muscle, skin, and thyroid tissues from the GTEx project database.

In the MH-only test, we took expression profiles for seven MH platforms (Table 1). Specifically, we took normal brain samples from the GTEx and TCGA datasets, and four specific model human mRNA sample types (called type A, B, C, D samples) from the MAQC and SEQC protect databases. Type A samples were the Stratagene Universal Human Reference RNA mixture for all but brain human tissues; type B samples were the the Ambion Human Brain Reference RNA. Type C samples were the mixture of A and B in the weight proportion 3:1, and Type D the mixture of A and B in the proportion 1:3.

In all these three tests, in the case of successful harmonization one can expect tissue type–specific rather than platform-specific clustering of the samples. In addition, in the MH-only test, the normal TCGA and GTEx brain samples should cluster together with sample B; sample B should cluster closer to sample D than to samples C and A, respectively. Likewise, sample A should cluster closer to sample C than to samples D, and B/brain samples, respectively.

Selection of auxiliary datasets Q and P for Shambhala harmonization:

Importantly, both linear (Shambhala-1) and cubic (Shambhala-2) methods require two datasets, P and Q , the latter acting as the reference for gene expression profiles after harmonization, and the former serving for preliminary calibration of expression level ranges. For both linear and cubic Shambhala modifications, we tested two different Q datasets, named Q_0 and Q_1 . Each Q -dataset contained 100 expression profiles for ten healthy human tissues: brain, nerve, skin, adipose, muscle, heart, lung, thyroid, blood vessels, and blood. For the auxiliary (calibration) dataset P , we tested eight different versions called P_0 – P_7 , accordingly. Except for the dataset P_2 , which represented mostly human cancer tissues, the samples represented healthy human tissues similar to the above list for Q -datasets (Table 2). Thus, we tested totally 32 different Shambhala modes:

2 (linear or cubic harmonization) · 2 (number of Q -datasets) · 8 (number of P -datasets).

Let a raw dataset (R) be a dataset to be harmonized; it includes N gene expression profiles obtained using different experimental platforms. The Shambhala-2 protocol (Fig. 1A) includes the following steps.

1. Take the first profile from the raw dataset and quantile-normalize it with the auxiliary calibration dataset P . This produces a dataset P' , where one of the samples is the transformed sample 1 ($1'$) from the initial dataset R .
2. Normalize the dataset P' using the CuBlock method (Junet et al., 2021), thus producing the dataset P'' including double transformed sample 1 ($1''$).
3. Take the sample $1''$ from the dataset P'' and rescale the expression levels for each gene g in accordance with the mean and the standard deviation of gene g expression level in the definitive dataset Q (μ_{gQ} and σ_{gQ} , respectively).
4. Repeat steps 1-3 for all other samples $2, \dots, N$ of the raw dataset; this results in the fully harmonized dataset of interest.

CAUTION: *We recommend that all Shambhala-2 harmonization procedures be performed in the logarithmic space.*

ALTERNATE PROTOCOL 1

LINEAR SHAMBHALA/SHAMBHALA-1

The first version of the Shambhala approach [or Shambhala-1 (Borisov et al., 2019)] used the transformation of dataset P'' into the shape of the dataset Q using the XPN method (Shabalin et al., 2008), which showed the best performance among the pairwise cross-platform normalization techniques (Rudy & Valafar, 2011). Namely, XPN [and, consequently, Shambhala-1 (Borisov et al., 2019)] employed stochastic clustering for genes and samples using genetic algorithms, and partially linear iterative harmonization of two datasets. However, the major distinction between XPN and Shambhala-1 (linear Shambhala) is that in the Shambhala-1 method, only one dataset (P'') is changed, while Q remains unchanged during the iteration steps, whereas in the XPN method both datasets are transformed iteratively.

Shambhala-1 was performed in R using the *HARMONY* package and is publicly available at <https://github.com/oncobox-admin/harmony>.

Materials

Same as for Shambhala-2 protocol (Tables 1 and 2)

Let a raw dataset be a dataset to be harmonized; it includes N gene expression profiles obtained using different experimental platforms. The Shambhala-2 protocol (Fig. 1B) includes the following steps.

1. Take a profile number 1 from the raw dataset, and quantile-normalize it with the auxiliary calibration dataset P . This produces a dataset P' with the transformed sample 1 from the dataset R ($1'$).
2. Convert it statically into the shape of the definitive dataset Q using the XPN method (Rudy & Valafar, 2011; Shabalin et al., 2008); this produces the dataset P'' with the double transformed sample 1 from the dataset R ($1''$).
3. Take the sample $1''$ from the dataset P'' .
4. Repeat steps 1-3 for all other samples $2, \dots, N$ of the raw dataset; this results in fully harmonized dataset of the interest.

CAUTION 1: We recommend that all Shambhala-1 harmonization procedures be performed in the logarithmic space.

CAUTION 2: For Shambhala-1 conversion, use spherical (cosine-based; Hornik, Feinerer, Kober, & Buchta, 2012; Krishna & Narasimha Murty, 1999) rather than barycentric (as in the XPN) clustering of samples in P' and Q datasets.

ALTERNATIVE (FLEXIBLE-FORMAT AND UNIFORMLY SHAPED) NORMALIZATION METHODS

ALTERNATE PROTOCOL 2

As an alternative to Shambhala-2 and for comparison purposes, we tested three other methods which produced output with flexible formats of expression data: quantile normalization (QN; Bolstad et al., 2003), DESeq2 (Love et al., 2014), and CuBlock (Junet et al., 2021), and three methods returning output data in uniform (pre-defined) formats: robust QN (QNR; Bolstad, 2008), TDM (Thompson et al., 2016), and UPC (Piccolo et al., 2013). Each of these normalizations was applied to pre-harmonized data and then compared with Shambhala-2 processed data.

Materials

Same as for Shambhala-2 protocol (Table 1)

1. Perform quantile normalization in R using the *normalize.quantiles* method from the *preprocessCore* package, publicly available at <https://github.com/bmbolstad/preprocessCore>.
2. Perform DESeq2 normalization in R using the *estimateSizeFactors* method from the DESeq2 package, publicly available at <https://bioconductor.org/packages/release/bioc/html/DESeq2.html>.
3. Perform CuBlock normalization using MATLAB, publicly available at <https://www.mathworks.com/matlabcentral/fileexchange/77882-cublock>. In its original form, CuBlock produces deviations from the mean level as output values. Thus, we rescaled the CuBlock output to make the mean and the standard deviation of the expression level of each gene g equal to μ_g and σ_g , respectively, i.e., to the mean value and standard deviation of the gene g expression level in the raw (pre-harmonization) dataset.
4. Similarly to the idea behind the fRMA method (McCall et al., 2010), i.e., to freeze a reference profile and quantile transform any input to that single reference, apply the *normalize.quantiles.robust* method from the *preprocessCore* R package (<https://github.com/bmbolstad/preprocessCore>) to normalize gene expression profiles according to QNR method. As a reference frozen dataset for QNR, we used the Q_0 -dataset to compare the results to Shambhala-2.
5. Perform TDM normalization using the *tdm_transform* method from the TDM R package (<https://github.com/greenelab/TDM>), with the target data being the dataset to be normalized and the refdata being the Q_0 -dataset, to compare the results with Shambhala-2.
6. Perform UPC normalization in R using the *UPC_Generic* method from the SCAN.UPC package, publicly available at <https://www.bioconductor.org/packages/release/bioc/html/SCAN.UPC.html>. After this, the UPC output was rescaled. Namely, we set the mean and standard deviation of the expression level of each gene g equal to μ_{gQ} and σ_{gQ} , respectively, i.e., to the mean value and standard deviation of gene g expression level in the Q_0 -dataset, to compare the results with Shambhala-2.

CAUTION: All normalization methods were applied in logarithmic space, except for the DESeq2, which operates in non-logarithmic (linear) space.

Borisov et al.

WATERMELON MULTISECTION (WM)

The following test was designed for the assessment of expression data harmonization quality. A successful harmonization is expected to result in the expression profiles that exhibit similarity primarily based on their biological features rather than on the technical platform used to interrogate gene expression. We therefore assessed whether clustering dendrograms tended to show biological similarity or experimental platform-based grouping of gene expression profiles after harmonization using different methods.

Due to its intuitive visualization, unsupervised nature, and deterministic outcome, hierarchical clustering is often the method of choice for the analysis of gene expression data. However, the results are most frequently manually curated based on the output graphical data, thus complicating its use for sophisticated tasks such as the analysis of large datasets and other poorly visualized objects. Recently, we developed a method termed Watermelon Multisection (WM; Zolotovskaia et al., 2020) designed to return a specific metric for the assessment of an entropy-based quality of clustering on dendrograms according to known classes. WM can automatically evaluate the performance of hierarchical clustering dendrograms relative to a trait of interest, e.g., known tissue type in our case. WM measures the strength of data association with the trait of interest, thus quantitatively annotating the entire clustering tree and its overall quality.

Here, we refer to the biological origin of samples as *sample class*, and the gene expression interrogation technique as *platform class*. When moving from the root of the dendrogram to its distal branches, one can measure entropy decrease and, therefore, information gain (IG) at each node of the dendrogram, i.e., its split into two shoulders (Zolotovskaia et al., 2020). In WM, the overall process of gradual information gain at each node is referred as the observed information gain trajectory. The WM metric equals the normalized difference between the observed IG; the theoretically maximal IG corresponds to the fastest separation of predefined classes into the distinct branches; and the null IG_0 trajectory describes the worst (totally random) distribution of predefined classes along the dendrogram (Zolotovskaia et al., 2020; also see Equation 1).

$$WM = \frac{IG_{observed} - IG_0}{IG_{ideal} - IG_0}$$

Equation 1

Therefore, $0 < WM < 1$. The lowest value here is observed when class labels are distributed randomly along the dendrogram, whereas the highest value is achieved if n classes are separated exactly during the first n dendrogram branching events (Zolotovskaia et al., 2020). In this application, the WM metric was applied for the assessment of harmonization quality in the following way:

- High-quality harmonization is expected to result in clustering according to biological origin, and not to the experimental platform or other technical factors. Thus, the ratio $R = \frac{WM_S}{WM_P}$ (Equation 2) of WM-metrics according to biological sample classes (WM_S) and experimental platform classes (WM_P) may be used for evaluating harmonization quality. The higher R is, the better harmonization quality.

Since the WM metric works more robustly for classes with a comparable number of samples, and also to accelerate the process of WM metric calculations for large datasets, we used the following protocol for WM metric evaluation. For each harmonizing dataset, we randomly selected five samples for each known combination of tissue type and experimental profiling platform. We then calculated the WM metrics for each selection. Each selection was randomized and repeated 25 times. In addition to WM-metrics, we also

used principal component analysis (PCA; Pearson, 1901), to visualize relative proximities of the harmonized molecular profiles.

NOTE: WM metric calculation code was implemented in R, publicly available at <https://gitlab.com/oncobox/cluster-analysis>.

Materials

We did three major harmonization quality tests with 8390 gene expression profiles corresponding to different human tissues and established using different equipment and/or reagents (Table 1). First, we assessed harmonization performance for five normal human tissue types profiled using three alternative RNAseq protocols, and extracted from the databases of the Genotype Tissue Expression (GTEx; GTEx Consortium, 2013), The Cancer Genome Atlas (TCGA; Tomczak et al., 2015), and the Oncobox Atlas of Normal Tissue Expression (ANTE; Suntsova et al., 2019) projects, for 2801 samples in total. In the second test, we evaluated harmonization of samples for five human tissue types profiled using both RNAseq and MH in the GTEx project, for 5140 samples total. Third, we compared clustering of human brain and non-brain samples profiled using different MH platforms in the Microarray Quality control (MAQC; Shi et al., 2006), Sequencing Quality Control (SEQC; Xu et al., 2016), GTEx, and TCGA projects, for 449 samples total.

These tests allowed us to assess harmonization quality for several RNAseq and MH platforms and for some of their combinations, and to estimate performance of Shambhala-2 in comparison to previous techniques.

1. For each harmonized dataset, randomly select five samples for each combination of tissue type and experimental profiling platform (i.e., if there are three tissue types and five platforms, there will be fifteen combinations in total—five random samples per each combination, thus making a total of 75 samples).
2. Calculate for this selection the WM metric for sample-type-based classes (WM_S) and for platform-based classes (WM_P).
3. Calculate the ratio $R = \frac{WM_S}{WM_P}$.
4. Repeat steps 1-3 25 times.

CALCULATION OF CANCER-TO-NORMAL LOG-FOLD CHANGE (LFC) AND PATHWAY ACTIVATION LEVEL (PAL)

In this study, we assessed capacities of different harmonization techniques to retain functional transcriptomic characteristics, such as the case-to-normal fold-change and the activation of molecular pathways. Sufficient retention of these metrics may suggest lack or loss of useful information in transcriptomic profiles after harmonization. To this end, we calculated the following gene expression-based values: cancer-to-normal log-fold change values ($LFCs$); and pathway activation levels ($PALs$; Borisov et al., 2020; Tkachev, Sorokin, Garazha, et al., 2020).

Materials

To evaluate retention of the above characteristics following harmonization with Shambhala-2, we extracted from the GTEx (GTEx Consortium, 2013), TCGA (Tomczak et al., 2015), and ANTE (Suntsova et al., 2019) databases 4086 cancer and the corresponding normal tissue samples profiled by RNAseq (Table 3).

For each cancer sample, calculate:

1. Log-fold change, LFC_n , i.e., $\log_2$ -ratio (Equation 4) of gene n mRNA concentrations in the test sample and in the norms (median value in the control group).

SUPPORT PROTOCOL 2

Borisov et al.

Table 3 Number of Samples for the Test of Differential Cancer/Normal Gene Expression Values and Their Physiological and Clinical Derivatives

Localization	Colorectum	Kidney	Lung
Cancer Oncobox	83	32	62
Cancer TCGA	672	903	1046
Normal Oncobox ANTE	6	6	7
Normal TCGA	51	129	110
Normal GTEx	507	45	427

2. Pathway p activation level, $PAL_p = \sum_n NII_{np} \cdot ARR_{np} \cdot LFC_n / \sum_n |ARR_n|$ (Equation 5). Here the node involvement index, NII_{np} , is the index of gene product n in relation to the pathway p , assuming the values equal to one for gene products included in the pathway and equal to zero for gene products not included in the pathway. The discrete value ARR_{np} (activator/repressor role) is determined for a gene n in the pathway p as follows:

$$ARR_{np} =$$

- 1; gene product n is a repressor of pathway p
- 0.5; gene product n is more a repressor than an activator of pathway p
- 0; unclear repressor or activator role in pathway p
- 0.5; gene product n is more an activator than a repressor of pathway p
- 1; gene product n is activator of pathway p

CAUTION: LFC and PAL values were calculated for all samples of each cancer type using available corresponding normal samples.

COMMENTARY

Background Information

We have updated a previous version of the Shambhala harmonizer of gene expression profiles (Shambhala-1) with the mathematical apparatus for cubic cross-platform harmonization that was published first for the CuBlock technique (Junet et al., 2021). We then compared performance of Shambhala-1 and Shambhala-2 in removing platform bias and in retaining tissue-specific gene expression features. In parallel, we assessed Shambhala-2's performance in comparison with six previous-generation harmonization techniques: three with flexible (non-uniform) output formats [quantile normalization (QN; Bolstad et al., 2003), DESeq2 (Love et al., 2014), and CuBlock (Junet et al., 2021)], and three with uniform (pre-defined) formats [(QNR; Bolstad, 2008), TDM (Thompson et al., 2016), and UPC (Piccolo et al., 2013)].

Critical Parameters

Shambhala2 converts gene expression profiles one-by-one, independently from one another, to a predefined shape that is set by the definitive dataset Q . The robustness of the re-

sults is strongly enhanced by using the auxiliary calibration dataset P .

Shambhala2 is released as an R function `Shambhala2 = function (InputFileName, PFileName, QFileName, delete_buffer_files = TRUE, k = 5)` that calls MATLAB applications `Shambhala2.m`, `CuBlock.m`, and `readExpressionData.m`. The parameter k is the number of probe clusters for the application of k -means to find probe-cluster partitions. The default setting of k is 5.

Troubleshooting

Table 4 lists problems that can arise with this procedure along with their possible causes and solutions.

Understanding Results

Comparison of Shambhala-2 (cubic harmonizer) to Shambhala-1 (linear harmonizer)

Both Shambhala protocols convert the gene expression profiles one-by-one into the format of a selected definitive reference dataset

Table 4 Troubleshooting Guide for Shambhala-2 Protocol

Problem	Possible cause	Solution
No <i>matrixStats</i> package in R	R requires <i>matrixStats</i> package	Type <code>install.packages("matrixStats")</code> at the R prompt
No <i>quantilenorm</i> method in MATLAB	MATLAB requires method <i>quantilenorm</i> (https://www.mathworks.com/help/bioinfo/ref/quantilenorm.html) from the <i>Bioinformatics</i> Toolbox (https://www.mathworks.com/help/bioinfo/index.html?s_tid=CRUX_lftnav)	Install the <i>Bioinformatics</i> Toolbox
I don't have MATLAB license	MATLAB license is free for educational institutions	Those who do not have access to MATLAB license may try to use GNU Octave instead

Table 5 Median (Over 25 Random Pickings) Values of $R = \frac{WM_S}{WM_P}$ for Three Harmonization Tests and Different *P*- and *Q*-Datasets for Linear and Cubic Shambhala Modes

	Test 1				Test 2				Test 3, Five Sample Classes				Test 3, Two Sample Classes			
	Q_0		Q_1		Q_0		Q_1		Q_0		Q_1		Q_0		Q_1	
	Linear	Cubic	Linear	Cubic	Linear	Cubic	Linear	Cubic	Linear	Cubic	Linear	Cubic	Linear	Cubic	Linear	Cubic
P_0	1.68	1.96	1.71	1.88	1.20	1.45	1.10	1.48	0.84	0.88	0.86	0.86	1.02	1.09	1.05	1.05
P_1	1.79	1.98	1.79	1.84	1.06	1.34	1.05	1.30	0.74	0.86	0.79	0.87	0.86	1.03	0.88	1.05
P_2	2.68	2.19	2.03	1.92	1.41	1.43	1.27	1.32	0.87	0.87	0.86	0.86	1.04	0.99	0.89	0.97
P_3	1.72	1.88	1.59	1.71	1.37	1.35	1.21	1.34	0.87	0.85	0.89	0.85	1.05	1.04	1.10	0.89
P_4	1.70	1.96	1.69	1.84	1.08	1.46	1.09	1.43	0.84	0.83	0.88	0.85	1.05	0.94	1.08	0.97
P_5	1.32	1.98	1.83	1.84	1.21	1.38	1.20	1.39	0.80	0.86	0.86	0.85	1.00	0.99	1.03	0.98
P_6	1.74	2.12	1.70	1.93	1.00	1.46	1.08	1.47	0.74	0.81	0.80	0.84	0.82	0.82	0.77	0.84
P_7	1.69	1.95	1.58	1.70	0.90	1.11	0.97	1.11	0.73	0.72	0.70	0.72	0.85	0.85	0.85	0.85

(*Q*). To support the robust convergence of harmonized profiles, they are converted within an auxiliary (calibration) dataset (*P*).

For each Shambhala mode (i.e., linear and cubic), we tested eight *P*-datasets (numbered from 0 to 7) and two *Q*-datasets (numbered 0 and 1). See Table 2.

For two out of three harmonization datasets, the cubic Shambhala significantly outperformed the linear mode (for most combinations of *P*- and *Q*-datasets, except for the dataset P_2 which had a limited number of genes): in Test 1 (RNAseq-profiles only; Table 5 and Supplementary Figs. 1-5) and in Test 2 (RNAseq and MH profiles; Table 5 and Supplementary Figs. 6-10). In Test 3, both modes showed similar effectiveness (MH-only, Table 5, Supplementary Figs. 11-16). See Supporting Information for all Supplementary Figures.

In Test 3, both linear and cubic Shambhala modes could not strictly distinguish between model sample types A and C, and between sample types B and D; however, it strongly supported clustering for the “*mostly brain*” (samples B, D, and “Normal brain”) and “*mostly non-brain*” (samples A, C) groups. We therefore performed Test 3 in two modifications: (i) with five classes of samples (A, B, C, D and Normal brain; Supplementary Fig. 11) and (ii) with two classes of samples (*mostly brain* and *mostly non-brain*; Supplementary Fig. 12).

Based on the results of Tests 1-3, we selected the best settings for Shambhala-2 (Sh2), which were “cubic mode, *P*-dataset 0, *Q*-dataset 0”. Such a mode allowed high-quality tissue-specific clustering of five tissue types in Tests 1 and 2 (Supplementary Fig. 3, Supplementary Fig. 8),

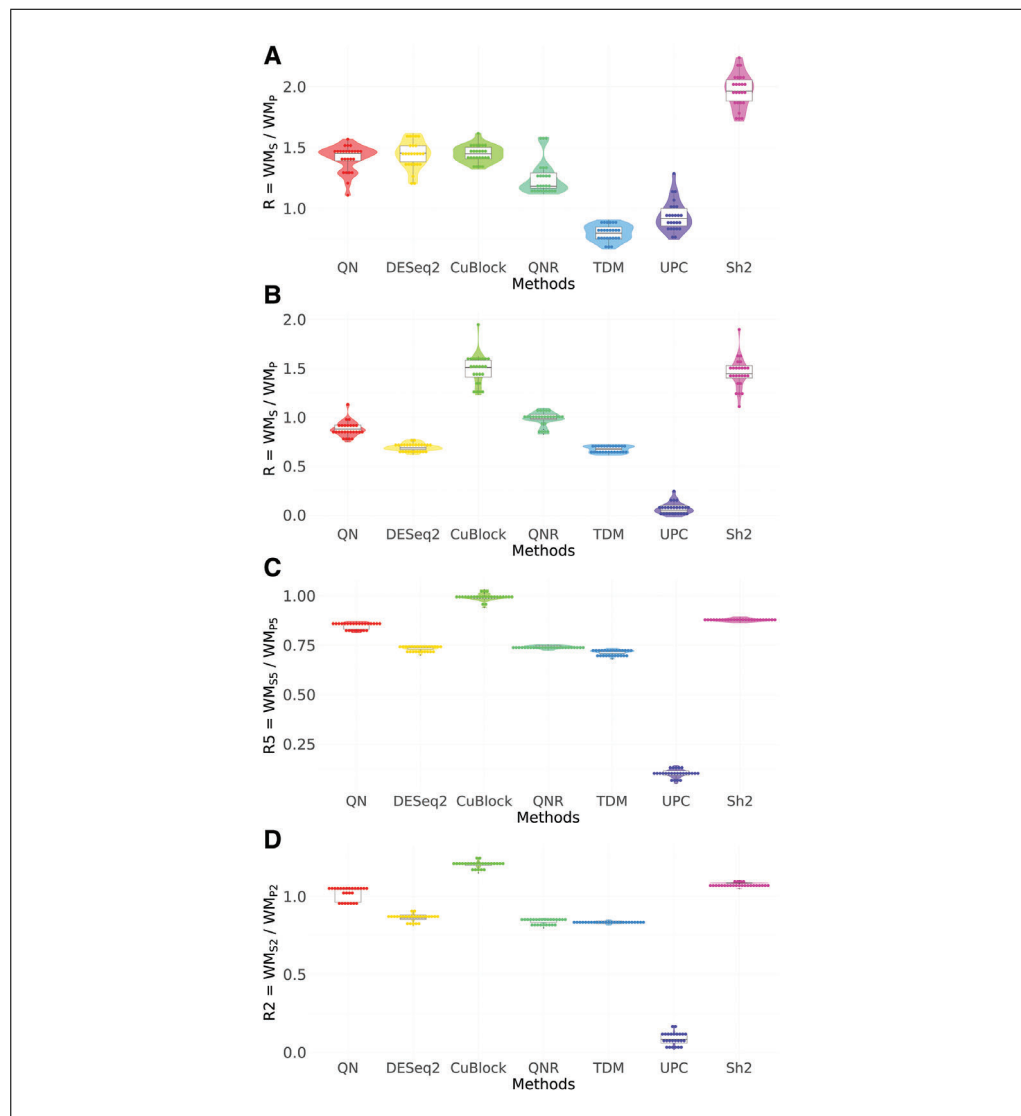


Figure 2 WM-metrics ranking for QN, DESeq2, CuBlock, QNR, TDM, UPC, and Shambhala-2 (Sh2) methods in Test 1 (A), Test 2 (B), Test 3, five sample types (C), Test 3, two sample types (D), and harmonization assay for two different classes (*Mostly brain*, and *Mostly non-brain*). On every panel, distribution of R , ratio of sample type–based (WM_s) to platform type–based (WM_p) metrics is shown.

and supported distinguishing “mostly brain” from “mostly non-brain” samples in Test 3 (Supplementary Fig. 14). In further tests, this version will be referred to as Shambhala-2 (Sh2).

Shambhala-2 comparison with previously existing normalization techniques

We then compared the capacity of Shambhala-2 to support sample type–specific clustering in a series of parallel experiments with the alternative harmonization techniques. To this end, in Tests 1-3 we did parallel harmonization using different methods: Shambhala-2 and several previously existing cross-platform normalization methods—three with flexible (non-predefined) output format

(quantile normalization (QN; Bolstad et al., 2003), Differential gene expression analysis based on the negative binomial distribution (DESeq2; Anders & Huber, 2010; Love et al., 2014), and CuBlock (Junet et al., 2021)—and three that produce normalized profiles in the uniformly-shaped (predefined) format: robust QN (QNR; Bolstad, 2008), Training Distribution Machine (TDM; Thompson et al., 2016) and Universal exPression Code (UPC; Piccolo et al., 2013).

To compare performances of these methods, we used the same Tests 1-3 as before (Figs. 2-5 and Supplementary Figs. 17-21). In terms of supporting tissue-specific clustering, Shambhala-2 outperformed QN and DESeq2 methods in all three tests

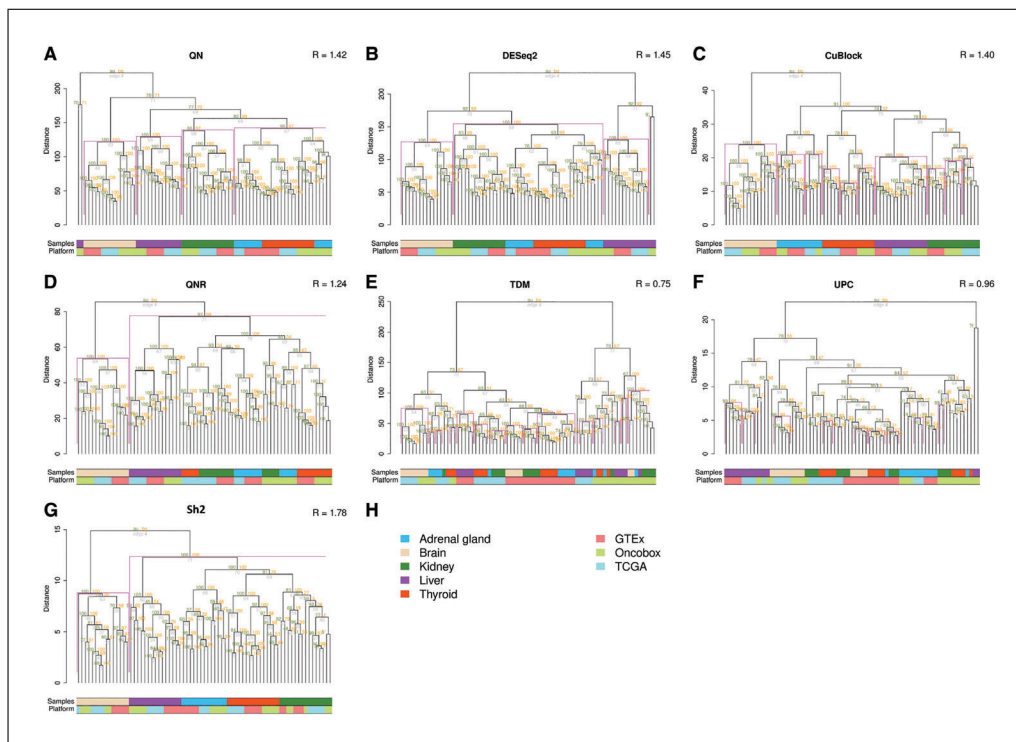


Figure 3 Cluster dendrograms for QN (A), DESeq2 (B), CuBlock (C), QNR (D), TDM (E), UPC (F), and Sh2 (G) for Test 1. Ratio of sample-based to platform-based WM-metrics (R) is shown on each panel in the upper right corner.

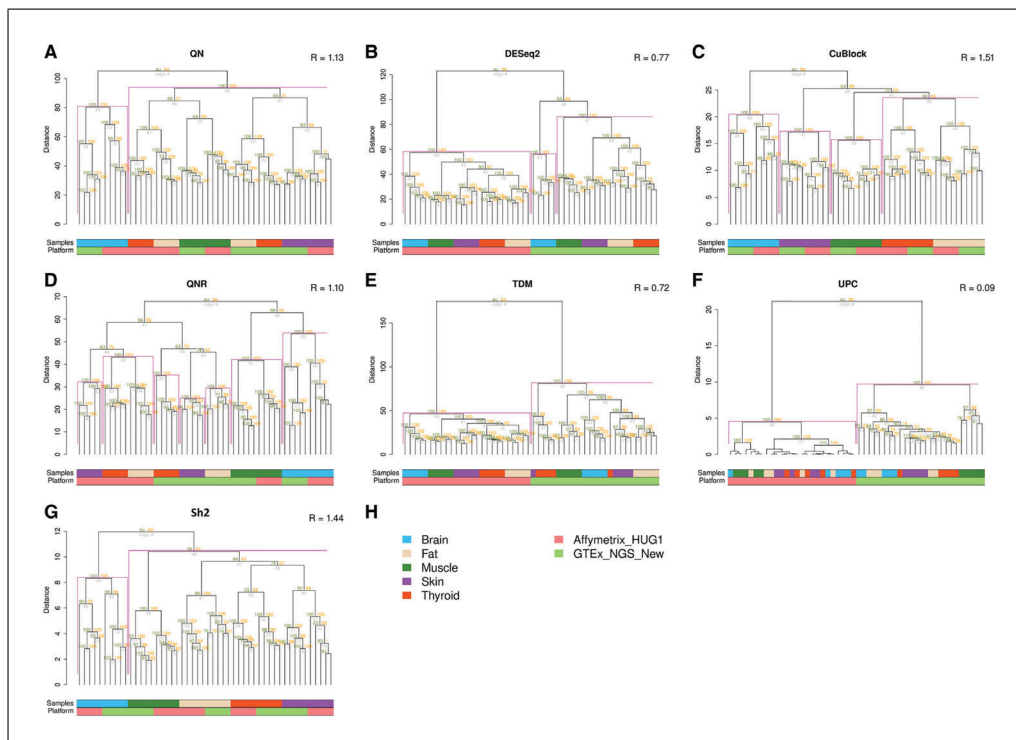


Figure 4 Cluster dendrograms for QN (A), DESeq2 (B), CuBlock (C), QNR (D), TDM (E), UPC (F), and Sh2 (G) for Test 2. Ratio of sample-based to platform-based WM-metrics (R) is shown on each panel in the upper right corner.

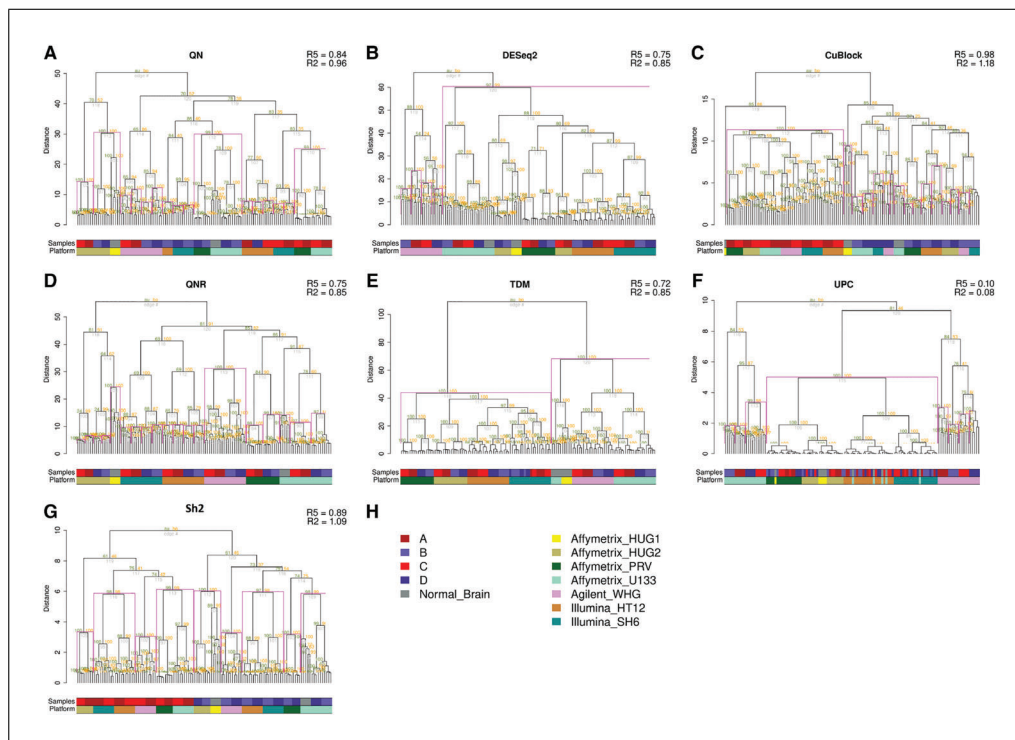


Figure 5 Cluster dendrograms for QN (A), DESeq2 (B), CuBlock (C), QNR (D), TDM (E), UPC (F), and Sh2 (G) for Test 3. R2 and R5, ratio of sample-based to platform-based WM-metrics for 2- and 5-sample groups, respectively.

(Figs. 2-5, Supplementary Figs. 17-21), as well as all three methods (QNR, TDM and UPC) with the uniform output. Interestingly, Shambhala-2 outperformed the CuBlock method in the RNAseq-only test (Test 1; Figs. 2A and 3, Supplementary Figs. 17-19), and showed comparable performance in the RNAseq versus MH test (Test 2; Figs. 2B and 4, Supplementary Figs. 17-18,20), but was less efficient in the MH-only test (Test 3; Figs. 2C,D and 5, Supplementary Figs. 17-18,21).

Shambhala-2 applicability to analyze cancer-to-normal log-fold-changes and pathway activation levels

In addition to retaining tissue-specific features after harmonization, it is also important to keep the differential expression patterns, which may play crucial roles in, e.g., oncology applications. Case-to-normal ratios are also used to calculate the pathway activation levels, which can be used as the predictive or diagnostic biomarkers (Aliper et al., 2017; Borisov et al., 2017, 2020; Buzdin et al., 2014, 2017, 2018). Furthermore, differential gene expression values in cancer versus normal tissues can also be used to calculate the predicted drug efficiency scores for target genes; such predictions can guide prescription of cancer drugs to the patients (Tkachev,

Sorokin, Garazha, et al., 2020; Poddubskaya et al., 2019; Zolotovskaia et al., 2019). Thus, we tested whether Shambhala-2 harmonization retained log-fold-change (LFC) and pathway activation level (PAL) values that correlated with those calculated for the pre-harmonization profiles.

To this end, we used RNAseq profiles for the colorectal, kidney, and lung cancers, and for the corresponding normal tissues obtained from the Oncobox ANTE (Suntsova et al., 2019), GTEx (GTEx Consortium, 2013), and TCGA (Tomczak et al., 2015) databases (see more details in Table 3). First, we took cancer tissue and the corresponding normal profiles from the Oncobox ANTE and TCGA databases (GTEx database was not included as it has no cancer profiles) and separately calculated cancer-to normal LFC and PAL values, both with and without Shambhala-2 harmonization. To normalize gene expression levels without Shambhala-2, we used the DESeq2 method, which is considered one of the most popular tools for normalization of RNAseq data (Maza, 2016; Varet, Brillet-Guéguen, Coppée, & Dillies, 2016). We then calculated correlations between the values obtained with and without Shambhala-2 harmonization (Fig. 6). We found that the harmonized gene counts, and both the LFC and PAL values exhibited relatively high correlations

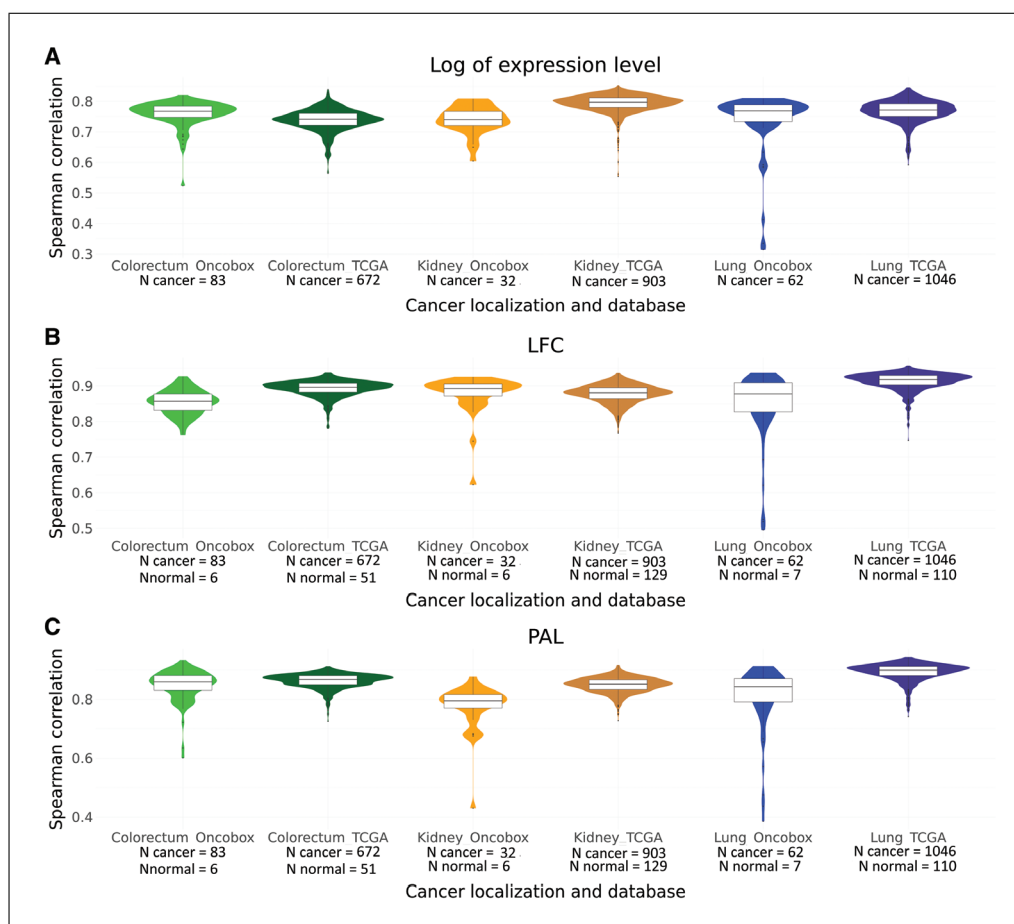


Figure 6 Spearman (rank) correlations between “with versus without Sh2 harmonization” modes for differential transcriptomic metrics in colorectal, kidney and lung cancers normalized on the corresponding healthy tissues. Log2 expression levels (A), LFC (B), and PAL (C).

(median 0.75 and higher, Fig. 6) in all the tests. Thus, we concluded that the differential transcriptomic metrics linearly correlated for the Shambhala-2-harmonized and for the initial RNAseq datasets.

Alternatively, we also tested whether Shambhala-2 can help when using unrelated normal datasets to compare cancer gene expression. Many pathologic tissue expression datasets lack the corresponding normal samples. Using unrelated normal datasets such as the GTEx collection can be problematic here, as such datasets could be obtained using different protocols and, therefore, platform-specific bias should be addressed. We investigated whether Shambhala-2 can be effective for crosslinking the unrelated cancer and normal tissue datasets. To this end, we normalized cancer and normal profiles using Shambhala-2 (“with Shambhala-2” mode) or using DESeq2 (“without Shambhala-2” mode). The normalized profiles were then used to calculate the LFC and PAL values for all possible combinations of cancer and corresponding normal profiles for the TCGA,

ANTE, and GTEx databases (Table 3). Finally, we correlated the transcriptomic metrics for all the scenarios when the same cancer type was normalized using its own norms, and norms from the unrelated datasets (Table 6 and Supplementary Figs. 23-28).

We found that in the majority of the comparisons using Shambhala-2 provided better results than using DESeq2 for merging cancer and unrelated normal datasets (Table 6). The advantage of using Shambhala-2 for harmonization of unrelated datasets in comparison with DESeq2 can be also observed on the corresponding PCA plots, both for the cancer and for the normal RNAseq profiles (Supplementary Figs. 28 and 29, respectively).

Discussion

Most of the algorithms for cross-platform normalization simultaneously transform all gene expression profiles within a dataset and provide flexible output formats. Consequently, flexibility of the obtained output formats leads to lower stability and reproducibility of the results. This means that one has to recalculate

Table 6 Median Spearman (Rank) Correlation Coefficients for LFC and PAL Values Calculated in Different Normal Samples With and Without Shambhala-2 Harmonization

			Colorectal cancer		Kidney cancer		Lung cancer	
			w/o Sh2	Sh2	w/o Sh2	Sh2	w/o Sh2	Sh2
Oncobox ANTE cancer dataset	ANTE norms versus TCGA norms	LFC	0.36	0.40	0.69	0.71	0.58	0.61
		PAL	0.29	0.49	0.80	0.87	0.59	0.73
	ANTE norms versus GTEx norms	LFC	0.64	0.70	0.72	0.74	0.67	0.71
		PAL	0.61	0.77	0.73	0.83	0.65	0.83
TCGA cancer dataset	TCGA norms versus ANTE norms	LFC	0.55	0.54	0.62	0.65	0.68	0.69
		PAL	0.75	0.74	0.80	0.88	0.87	0.86
	TCGA norms versus GTEx norms	LFC	0.60	0.58	0.63	0.63	0.76	0.75
		PAL	0.77	0.72	0.74	0.85	0.89	0.90

all the results after updating initial dataset(s) by new expression profiles.

On the other hand, the examples when the output format is predefined by a reference standard, and, therefore, stable, are rare (McCall et al., 2010; Thompson et al., 2016; Piccolo et al., 2013; Bolstad, 2008). In addition, the cases when cross-platform normalization is applied to merge together MH and RNAseq gene expression profiles are even more infrequent (Thompson et al., 2016; Piccolo et al., 2013; Tang et al., 2021; Xu et al., 2016; Angel et al., 2020).

Previously, we proposed the concept of one-by-one independent harmonization for each gene expression profile (Borisov et al., 2019). The key feature of the Shambhala (both linear and cubic) universal harmonization approach is converting transcriptomic data into a gold standard reference format, which is maintained by the samples present in the definitive dataset *Q*. Each of the profiles under harmonization is processed within the auxiliary dataset *P*, but separately from the other experimental profiles.

More recently, Junet and colleagues published a successful method of cubic cross-platform normalization that was specially designed for harmonization of different types of MH profiles (Junet et al., 2021). We have now adapted the advantage of cubic cross-platform normalization to improve the Shambhala universal harmonization scheme. In this study, we have tested the independent harmonization approach for both previous linear and new cubic modes by trying two different *Q*- and eight al-

ternative *P*-datasets. We performed three types of functional tests, one for the comparison of RNAseq platforms, another for the RNAseq versus MH platforms, and the third for different MH platforms. In each type of computational experiment, we used five different tissue types or sample types which, however, exhibited strong platform-specific bias before the harmonization.

As a quantitative criterion of harmonization quality, we used the Watermelon Multisection (WM) metric (Zolotovskaia et al., 2020). This helped to measure quality of clustering on dendrograms by either sample type or by the type of experimental platform used. We used the ratio of tissue-based versus platform-based WM metrics to assess harmonization quality.

The results showed that in most applications, the cubic harmonization mode indeed outperformed the linear mode. In all tests, Shambhala-2 worked better than flexible-format QN and DESeq2 methods, as well as the three stable-format normalization methods [robust QN (QNR; Bolstad, 2008), Training Distribution Machine (TDM; Thompson et al., 2016), and Universal exPression Code (UPC; Piccolo et al., 2013)]. In RNAseq-only tests, Shambhala-2 outperformed the ancestral CuBlock method. In RNAseq-versus-MH tests, Shambhala-2 and CuBlock showed comparable results. Finally, in the MH-only test, the CuBlock method showed superior performance to Shambhala-2. Thus, Shambhala-2, which is a harmonizer with the universal output format, showed best in its class or comparable-to-superior performance with

relation to most current stable and flexible harmonizer tools.

Thus, we validated the method for universal harmonization of transcriptomic profiles that enables comparison of multiple samples obtained in different datasets, using different experimental platforms and reagent settings. To our knowledge, this is the most successful approach for co-harmonization of RNAseq and microarray data, compared to TDM and UPC, which had been previously used for such co-harmonization (Thompson et al., 2016; Piccolo et al., 2012).

Furthermore, we showed that the Shambhala-2 harmonization retains functional transcriptomic trends, as evidenced by high correlations between pre- and post-harmonization gene expression fold-changes and pathway activation levels. In addition, it was also found useful for another task—merging pathological and normal tissue expression datasets—as exemplified here by crosslinking the unrelated cancer and normal tissue RNAseq datasets (Table 6).

Thus, we hope that Shambhala-2 or its daughter approaches will become useful for the broad spectrum of applications centered around combining different expression datasets. One of its major strengths is stability of final output data, which are not biased/distorted upon each new round of harmonization. Thus, it may become valuable for finding condition-specific features, e.g., for the assessment of tumor-specific gene expression or pathway activation patterns (Buzdin et al., 2020, 2021). This type of analysis will be especially important when the experimental (case) and control samples come from the different datasets, which is a common situation in human physiology research.

Another direction could be creating universal data collections of human transcriptome profiles from multiple source datasets, which might enable classifying samples by similarity of their harmonized gene expression profiles. Finally, the same approach can be transferred to non-human objects as well, which may require generating new adequate *P*- and *Q*-datasets and technical tuning of the method.

Time Considerations

Note that execution of the Shambhala-2 code may be time-consuming due to one-by-one separate harmonization of every profile. For example, processing the file `Input.csv` (<https://github.com/BorisovNM/Shambhala2>) takes about 2 min on an Ubuntu 20.04.2 LTS machine with Intel® Core™ i7-

7700K CPU @ 4.20 GHz × 8 CPUs and 31.2 GB or RAM. Therefore, parallel execution of the code may be advantageous.

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Author Contributions

Nicolas Borisov: designed the overall research, performed all harmonization and quality metrics calculations, and wrote the manuscript; **Maksim Sorokin:** suggested and pre-processed the datasets for the harmonization tests; **Marianna Zolotovskaya:** implemented WM scoring and performed the PCA; **Constantin Borisov:** pre-processed expression datasets and prepared figures; **Anton Buzdin:** supervised and designed the research and wrote the manuscript. All authors have read and approved the manuscript.

Conflict of Interest

M.S., M.Z., and A.B. were employed by Omicsway Corp. Walnut, Calif. M.Z. was also employed by Oncobox Ltd., Moscow, Russia.

Other authors declare no potential conflict of interest with the commercial entities.

Data Availability Statement

Q- and *P*-datasets used for linear and cubic harmonization, the datasets for harmonization during Tests 1, 2, 3, the cancer and the corresponding normal tissue gene expression dataset, and the calculated LFC and PAL for all cancer samples under investigation are deposited in the Zenodo repository <https://zenodo.org/record/6415067>.

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